Early Warning of Pitcher Arm Injuries from Pitch-Tracking Time Series (MLB Statcast)

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Motivation

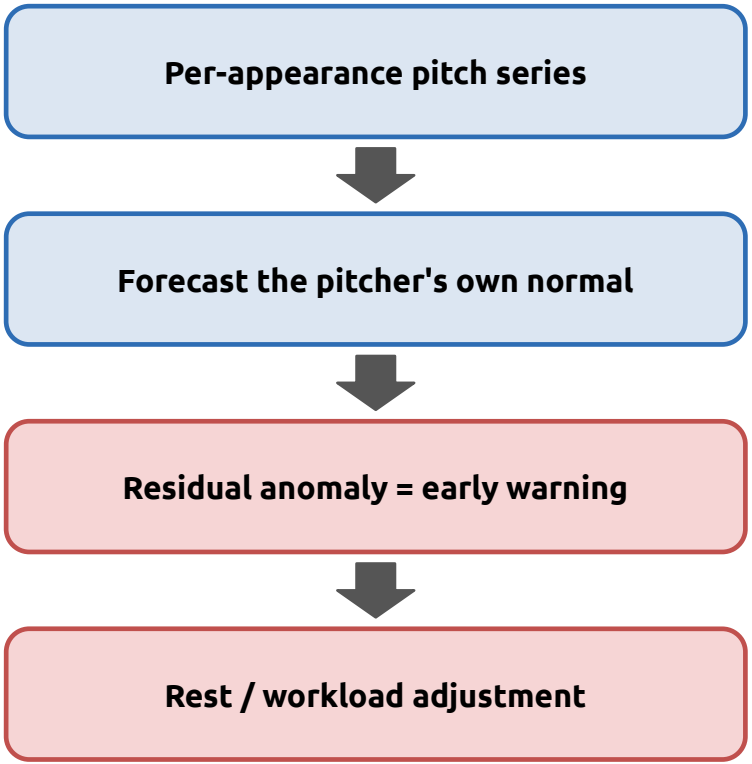
Why pitcher injury matters

- **Pitcher injuries keep rising**
  - MLB pitcher IL placements: **212 (2005) → 485 (2024)**; IL days **13,666 → 32,257** (AP, 2024)
  - Elbow-related IL days roughly **tripled**: 3,940 → 12,185
- **The cost is large**
  - 2015–2025: **1,122** pitchers, **2,365** shoulder/elbow IL placements, about **$3.33B** in lost salary; **68.3%** from elbow injuries (Dillon et al., OJSM)
- **Why it matters**
  - Once a pitcher is on the IL, the damage is already done
  - If warning signs are caught early → adjust rest days, manage workload, give rest

Motivation

Injury risk is a time series

- **Each pitcher has his own normal**
  - Velocity, velocity variability, release point per appearance
  - **Trend**: league-wide "velocity revolution" (velocity rises every season)
  - **Seasonality**: velocity tends to rise later in the season
  - **Exogenous drivers**: rest days, recent pitch counts (7/14/28 days), ACWR, temperature
- **If we can forecast a pitcher's normal, deviations become warnings**
- **Connects to our course**
  - ARIMA / ARIMAX, EWMA, anomaly detection on residuals



Related work

(1) Velocity as an injury signal

Study	Research Focus	Methodology	Important Contribution	Limitation
Prodromo et al. (2016)	Pitch characteristics before Tommy John surgery (UCLR) vs. age-matched controls	Case-control, conditional logistic regression	Higher fastball velocity before surgery; velocity is a significant variable	Compares pitcher vs. pitcher with season averages; cannot tell when change happens
Mayo et al. (2021)	Trends in the last 15 games before UCLR (223 pitchers)	Mann-Kendall trend test	Four-seam velocity fell ~0.1 mph/game (tau = −0.657, P < .001); spin rate fell ~5 rpm/game	No healthy controls; no alert rule
Mastroianni et al. (2026)	Pitch metrics before IL vs. healthy baseline	Linear mixed models with personal baseline	Last 5 appearances: velocity drops (P = .019), horizontal release variability rises (P = .005)	Retrospective comparison; no online, updating detection

Related work

(2) Detection methods and workload

Study	Research Focus	Methodology	Important Contribution	Limitation
Ozaki (Master's thesis, AUT)	Detecting deviation before UCLR (46 pitchers, 2016–2024)	Autoencoder anomaly detection on 40 features	Deviation grew 2.7× near surgery; code and matched controls (45 pairs) are public	Healthy pitchers also flagged (73% vs. 66%); intentionally ignores time order
Bright et al. (2026)	Injury-risk screening from broadcast video	Time-aware ML with irregular-interval correction	Personal baseline + interval weighting improves prediction	Uses video estimates, not measured tracking data
Slowik et al. (2021)	Workload and upper-extremity injury in college pitchers (n = 17)	ACWR with ROC threshold	ACWR change beyond ±33% → 8.3× next-week injury risk	Tiny sample, wide CI (1.8–54.1); 3-week chronic window

Research Background

Limitations of Previous Studies and Novelty (1)

- **Limitations of Previous Studies**
  - Most studies compare injured vs. healthy pitchers with season averages; within-pitcher change over time is rarely modeled.

➡ Injury signals should be read against each pitcher's own time series.

  - Pre-injury trends were tested, but never turned into an alert rule; false alarms on healthy pitchers were not counted.

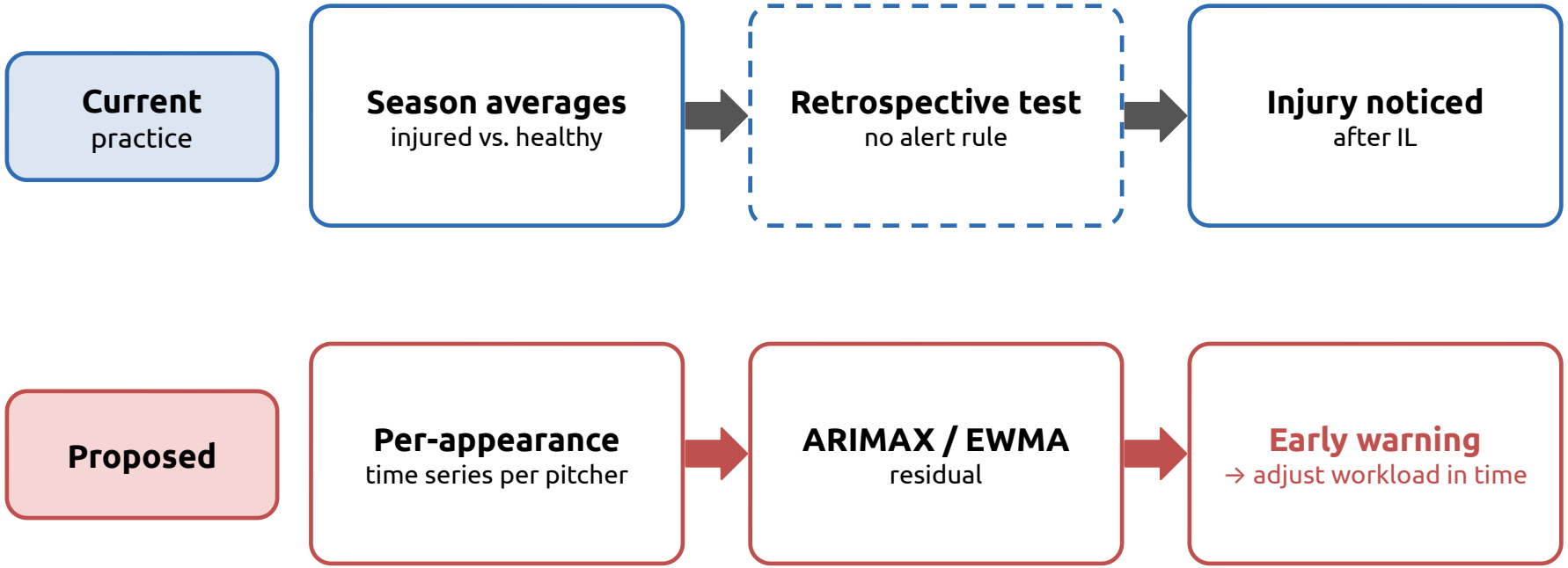
➡ We need an early-warning system evaluated by lead time and false alarms.

  - The one anomaly-detection study avoided time-series models, fearing they absorb slow decline into the baseline.

- **Novelty of the Proposed Method**
  - Pitcher-specific time-series baseline: forecast each pitcher's expected velocity, flag residuals
  - Performance + workload together: velocity, variability, release + rest days, 7/14/28-day counts, ACWR
  - Interpretable early-warning evaluation on open data: lead time, false alarms, healthy controls

Research Background

Limitations of Previous Studies and Novelty (2)



Dataset: MLB Statcast (2021–2026)

Dataset

- **Sources**
  - Pitch data: Baseball Savant **Statcast**, pitch-by-pitch
  - Injury labels: **MLB official API transactions** (IL placement date, body part)
  - Temperature: MLB API game weather, filled with Open-Meteo if needed
- **Data info**
  - Period: **2021–2026** regular seasons
  - Size: ~**4,500** pitches/day, ~**700K** pitches/season; ~**250** arm-injury events/season
  - Frequency: pitches aggregated per appearance; starters every ~5 days, irregular
  - Key variables: velocity, release side/height, extension, spin rate, pitch type, pitcher, date, game ID
  - Exogenous: rest days, 7/14/28-day pitch counts, temperature, month, starter vs. reliever
- **Research suitability: long, open, pitch-level series linked to real injury labels, so warnings can be tested against what actually happened.**

Dataset: MLB Statcast (2021–2026)

Unit of analysis and variables

Group	Variable	Definition
Performance	**Velocity**	Mean & max fastball velocity per appearance
	Velocity variability	SD of fastball velocity per appearance
	Release point	Mean & SD of horizontal / vertical release
	Auxiliary	Spin rate, extension, in-game velocity decay slope
Workload	**Rest days**	Days since last appearance
	Recent load	7/14/28-day cumulative pitches
	ACWR	7-day ÷ 28-day average load

- **Unit**: pitcher × appearance; time axis = appearance index, rest days as a separate variable
- **Injury event**: IL placement for an arm-related reason
- IL records are used only for evaluation, never for training

Injury early warning

Approaches (1)

- **Approach: early warning as time-series anomaly detection**
  - Learn each pitcher's normal level from his past appearances; score how far the current appearance deviates
- **Research questions**
  - **RQ1.** Do velocity, its variability, and release point move away from a pitcher's own normal in the last few appearances before injury?
  - **RQ2.** How early and how accurately can the residual (actual – forecast velocity) detect injury?
  - **RQ3.** Do workload indicators (rest days, 7/14/28-day pitch counts) improve alert performance?
  - **RQ4.** Does signal strength differ for starters vs. relievers and hard throwers vs. others?
- **Alert success**
  - Alert raised within the last K appearances before IL placement (K = 5 candidate, following Mastroianni et al.)

Injury early warning

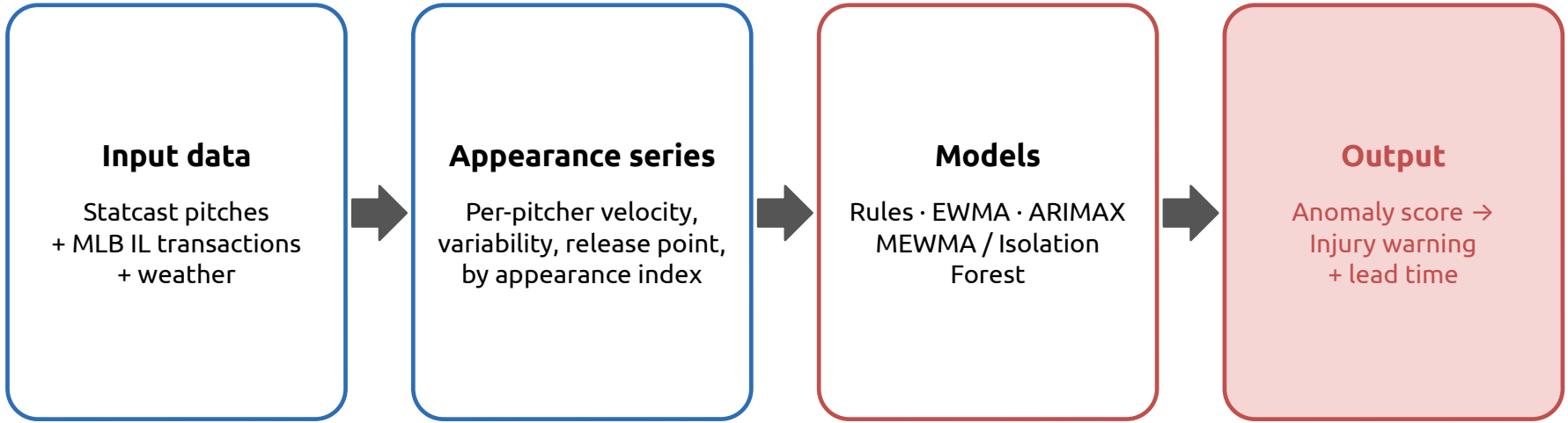
Approaches (2)

- **Model Selection**
  - Baseline 1: Random score (chance level)
  - Baseline 2: Simple rule (velocity drops X mph below season average)
  - Baseline 3: Workload rule (ACWR outside ±33%)
  - Model 4: **EWMA** (cumulative detection of gradual standardized velocity drop)
  - **Model 5: ARIMAX** (expected velocity given rest days, temperature, ballpark; anomaly = residual)
  - Model 6: **Multivariate** (MEWMA, Isolation Forest): velocity + variability + release + workload
  - (Optional) LSTM autoencoder
- **Evaluation**
  - Time-ordered split: train **2021–2024**, validation **2025** (alert threshold), test **2026**
  - Subgroups: hard throwers vs. others, starters vs. relievers

Task	Metric
Velocity prediction	MAE, RMSE
Injury detection	F1
Classification	AUROC, AUPRC
Early warning	Lead time (+ false alarms on healthy pitchers)

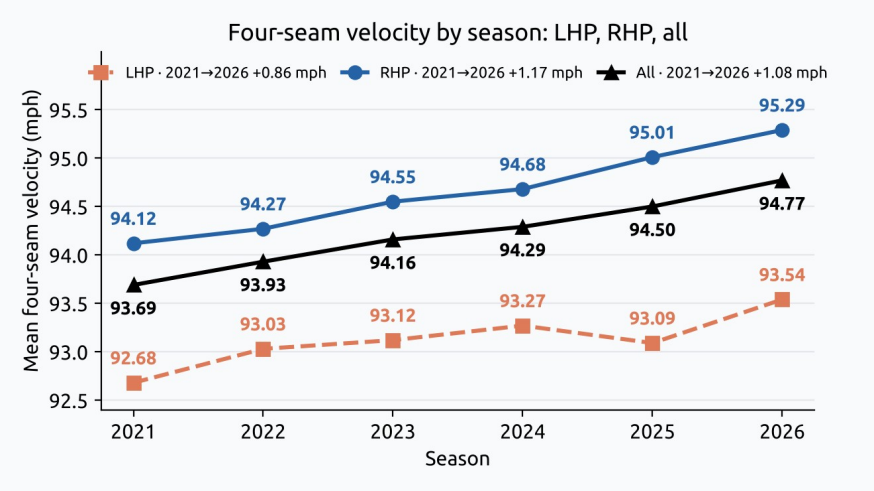
Injury early warning

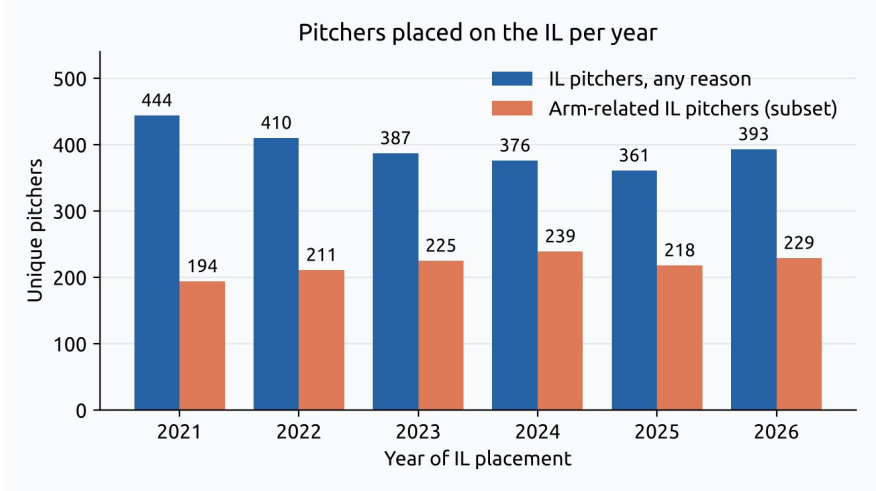
Approaches (3): framework



EDA of the dataset

Exploratory Data Analysis (1)



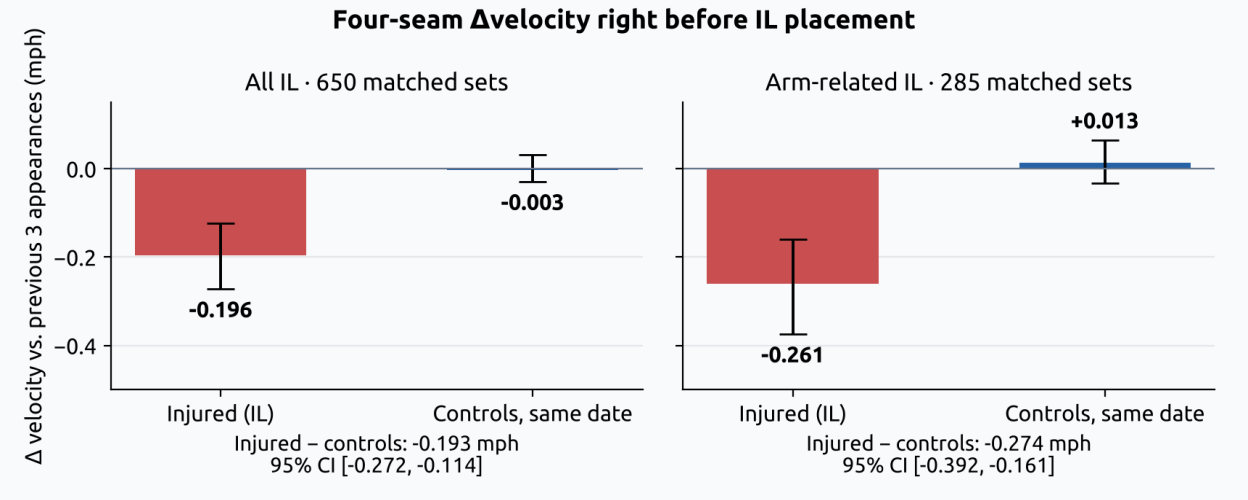


Annual four-seam velocity (left) and IL pitchers per year (right), 2021–2026 (own analysis)

- **Upward trend**: league four-seam velocity **93.69 → 94.77 mph** (+1.08) in 2021–2026; RHP +1.17, LHP +0.86 → need a pitcher-specific, detrended baseline
- **Arm injuries are a large share**: arm-related IL pitchers **194–239 per year**, about **44% (2021) → 58–64% (2023–2026)** of all IL pitchers

EDA of the dataset

Exploratory Data Analysis (2)

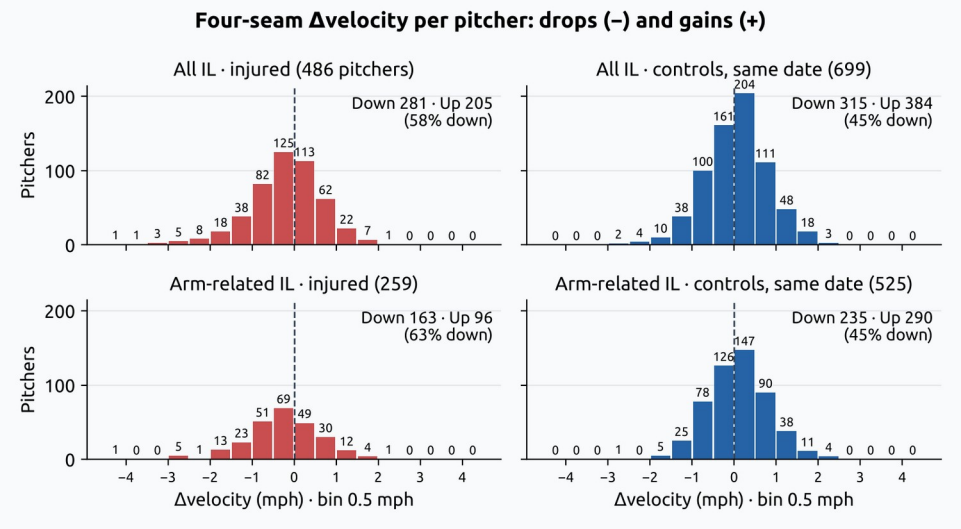


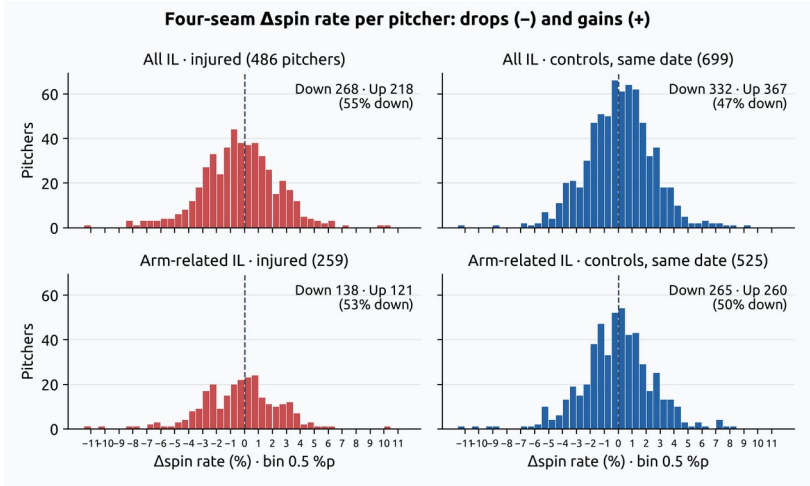
Four-seam Δvelocity right before IL: injured vs. controls at the same date (bootstrap 2,000×)

- Δvelocity = last appearance vs. average of previous 3 appearances
- All IL: **–0.196 mph** vs. controls –0.003 → difference **–0.193** [95% CI –0.272, –0.114]
- Arm-related IL: **–0.261 mph** vs. controls +0.013 → difference **–0.274** [95% CI –0.392, –0.161]
  → a measurable pre-injury velocity drop

EDA of the dataset

Exploratory Data Analysis (3)





Distribution of four-seam Δvelocity (left) and Δspin rate (right): IL vs. controls (own analysis)

- **Velocity**: arm-related IL **163 down / 96 up (63% down)** vs. controls 235 / 290 (45% down)
- **Spin rate**: arm-related IL 138 / 121 (53% down) vs. controls 265 / 260 (50% down) → weak signal, auxiliary only

➡ Distributions overlap heavily: a single-feature threshold will cause many false alarms → combine velocity, variability, release, and workload; evaluate false alarms

Expected Contributions

Expected Contributions

Individualized Modeling

Anomaly detection against each pitcher's own velocity level

Temporal Pattern Analysis

How velocity changes and how long anomalies persist before injury

Workload Integration

Rest days and ACWR as exogenous drivers

Early Injury Detection

Test whether warnings come before IL, with lead time and false-alarm rate